



MACHINE LEARNING PIPELINE

Preprocessing, Post-Processing & Evaluation

A practical implementation framework for building models that are efficient, interpretable, and reliably generalizable.



Preprocessing



Post-Processing



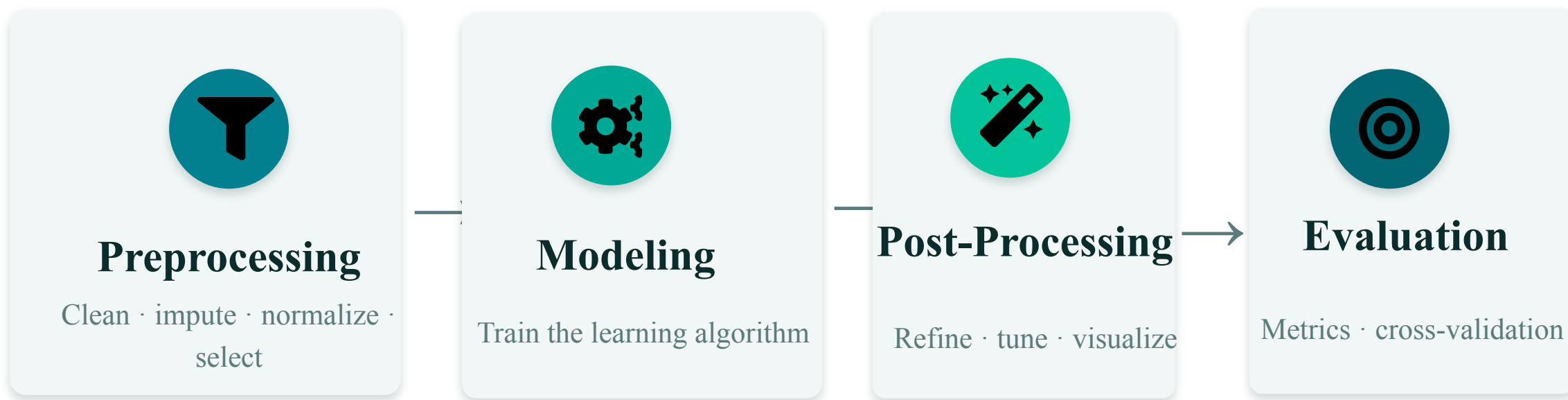
Evaluation

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The End-to-End Pipeline

Three coordinated stages turn raw data into a trustworthy, production-ready model.



Goal: maximize model efficiency and interpretability while ensuring the result is robust and generalizes beyond the training data.



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Data Preprocessing

Foundation steps that improve model efficiency before training even begins.



Data Cleaning

Remove duplicates, correct inconsistent entries, and filter noisy or invalid records.



Handling Missing Values

Impute gaps via mean/median/mode or model-based estimation, or remove sparse rows.



Normalization

Rescale features to a common range so no variable dominates due to its scale.



Feature Selection

Retain the most informative variables to reduce dimensionality and overfitting.

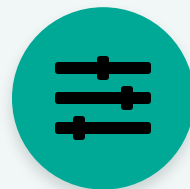
Post-Processing Techniques

Better interpretation of model outputs after inference.



Result Refinement

Smooth, filter, or aggregate raw predictions to remove noise and improve consistency before they're reported or acted upon.



Threshold Tuning

Adjust decision boundaries on predicted probabilities to balance false positives against false negatives for the task at hand.



Visualization

Turn outputs into confusion matrices, ROC curves, and dashboards so stakeholders can interpret results at a glance.

Post-processing bridges raw model output and human decision-making.

Evaluation Metrics

Quantifying how well predictions match reality.



Accuracy

$$(TP+TN) / \text{Total}$$

Share of all predictions the model got right.



Precision

$$TP / (TP+FP)$$

Of predicted positives, how many were correct.



Recall

$$TP / (TP+FN)$$

Of actual positives, how many were found.



F1-Score

$$2 \times P \times R / (P+R)$$

Harmonic mean balancing precision and recall.

TP = True Positive · TN = True Negative · FP = False Positive · FN = False Negative

No single metric tells the full story — choose the combination that matches the cost of errors in your problem.

Validation for Robustness

Confirming the model generalizes beyond the data it was trained on.



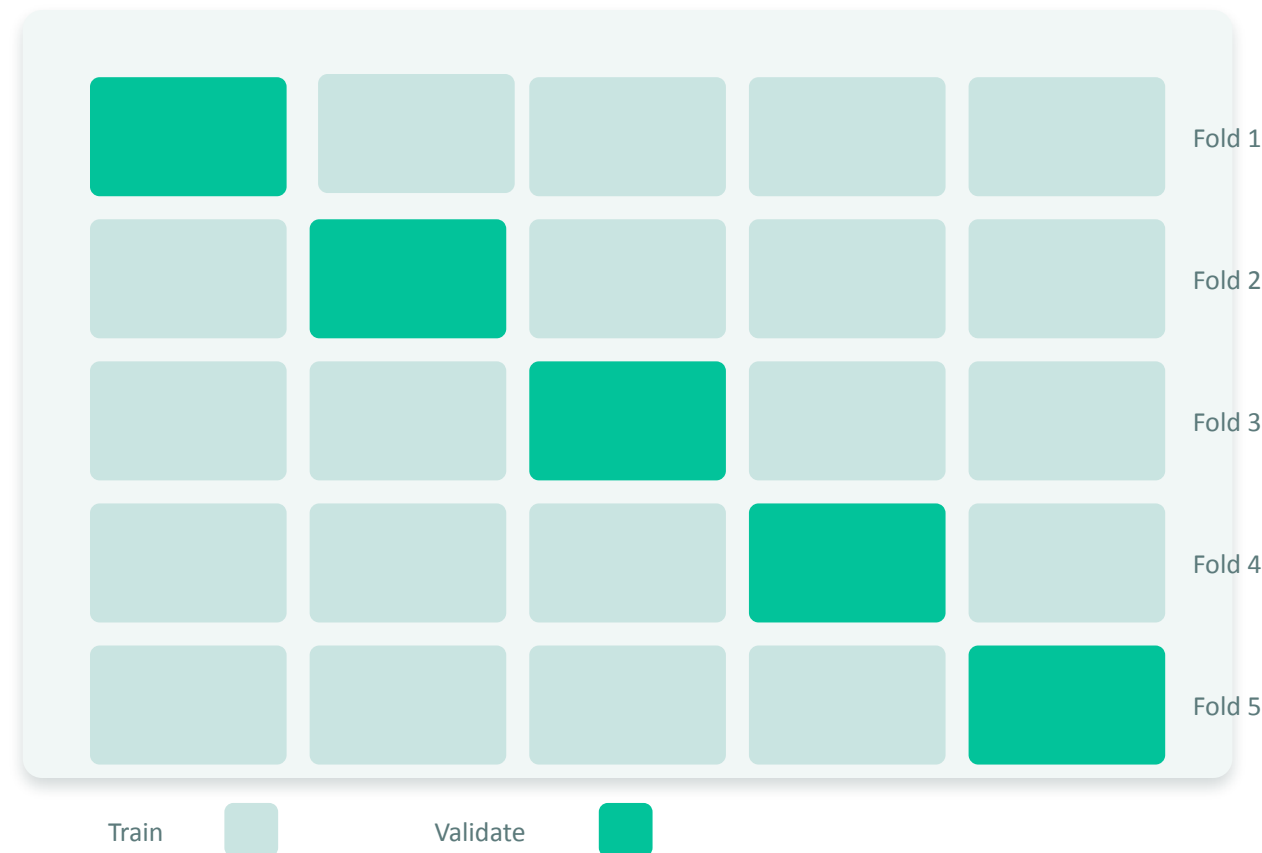
K-Fold Cross-Validation

The dataset is split into k equal folds. The model trains on $k-1$ folds and validates on the remaining fold, rotating until every fold has served as the validation set.

Averaging performance across all folds gives a more reliable estimate of real-world accuracy than a single train/test split — reducing the risk that results are a lucky (or unlucky) fluke.



Result: a robust, generalizable model ready for deployment.



Key Takeaways



Preprocessing (cleaning, imputation, normalization, feature selection) sets up an efficient, well-conditioned model.



Post-processing (refinement, threshold tuning, visualization) makes model output actionable and interpretable.



Evaluation metrics — accuracy, precision, recall, F1 — must be paired with cross-validation to confirm robustness.

Together, these stages form a disciplined pipeline for models that perform well — and keep performing well.

THANK YOU